

STAGE 4 TECHNOLOGY

Farm in a Box

Grow more with less

Student workbook · Edition 1.0



An illustrative system. Your crop, container and growing method require teacher approval.

Name _____

Class _____ Teacher _____

FRONT

Your learning route

Explore how agricultural systems work, then design, establish and improve a small crop-production system. Use the readings to prepare, the activities to explain your thinking, and the folio to keep evidence of your own decisions.

- 01 Agriculture Is Everywhere · pages 4–7
- 02 Country, Care and Future Farms · pages 8–11
- 03 Start a Living Crop · pages 12–15
- 04 Soil, Media and Circular Nutrients · pages 16–19
- 05 Water-Smart Growing · pages 20–23
- 06 Farm Under Pressure · pages 24–27
- 07 Design the Farm in a Box · pages 28–31
- 08 Build and Establish · pages 32–35
- 09 Grow, Measure and Improve · pages 36–39
- 10 Harvest, Judge and Share · pages 40–43
- Project folio · pages 44–50

Read before you respond. Record real observations, including missing data and unsuccessful tests. A crop that grows slowly or fails can still provide strong evidence when you explain what happened and propose a sensible next step.

50 A4 pages · 25 sheets when printed double-sided, long-edge binding. Your teacher may use the course website and slides alongside this book.

FRONT

The design challenge

Design, establish, monitor, improve and evaluate a portable Farm in a Box crop-production system using the crop, location, materials, tools and processes approved by your teacher. Use evidence to show how the system supports plant growth, manages resources and could be improved for another growing cycle.

Your evidence includes a defined need and measurable criteria, crop research, two genuinely different annotated concepts, a justified design, a production plan, safe establishment, dated observations, a test or modification, and a final judgement.

Turn a broad aim into a check

Broad aim: "Easy to care for."

Checkable criterion: "The approved care routine can be completed during the available visits, with access to observe the roots and water pathway."

Evidence: record the care steps, time needed and any access problem.

The teacher confirms the actual care arrangements. This worked example does not set a required time limit.

Before practical work, confirm the crop, location, materials, budget, tools, protective equipment and local controls with your teacher. This book does not approve pesticides, fertiliser recipes, mains-powered equipment or eating the harvest. Formal assessment dates and marks come from your teacher.

Write one question your team must resolve before selecting its design. Explain why the answer matters.

MODULE 01

Agriculture is everywhere

Agriculture connects everyday life



Figure 1. Finished products can hide their agricultural beginnings. Trace each one back to a living source.

A loaf of bread and a cotton shirt look very different, yet both begin with living plants. Agriculture organises the production of useful plants, animals and other biological products. People, land, water, knowledge and equipment work together. The farm is one part of a larger production system that also includes processing, transport and markets.

Four words help explain the connections. Inputs are resources brought into the system, such as seed, water, labour and energy. Processes are actions such as growing, caring for and harvesting. Outputs include the wanted product, but also waste, observations and experience. Feedback is information used to improve a decision. A growth record becomes feedback

when a producer uses it to change the next action.

Two NSW industries show why context matters. Riverina–Murray rice production supplies edible grain that is harvested and milled. Seasonal conditions, stored water and irrigation allocation influence production. NSW cotton supplies long plant fibres, separated from seed for textile manufacture, and has an expanding southern NSW presence. Water availability and seasonal conditions matter here too. Grain and fibre have different properties, so their processing and quality decisions differ. Agriculture also includes livestock and horticulture: no single production story represents them all.

Trace a cotton shirt backwards: identify its living source, one process and one example of feedback.

MODULE 01

From paddock to product

Follow the product, find the decisions

An agricultural product moves through connected stages before reaching a user. A useful starting model is grow or raise, harvest, transform, package, distribute and use. This is a thinking tool, not a fixed recipe. Some products repeat stages or do not need every stage. Drawing arrows between verbs helps show what changes, rather than just listing places.

Harvesting separates a useful product from its growing setting. Transformation changes its form or properties: milling grain into flour is one example. Packaging may protect and contain a product, carry handling information and make transport easier. Distribution moves products and information towards users. Each stage can

add value through usefulness, quality, access or information.

Value is only half the story. A stage can also use water, energy, materials and labour, or create damaged product and waste. Imagine comparing two packaging ideas. Removing all packaging might reduce material use but increase damage. The stronger decision would preserve protection while testing whether avoidable material or product loss falls.

Traceability means keeping reliable records that allow a product or batch to be followed. Records help locate a quality problem and its possible causes. A picture on a packet is not enough to prove a product's complete history; evidence must connect the stages.



Figure 2. A product changes as it moves. Ask what value, resources and evidence appear at each stage.

Choose packaging or distribution. Explain one value it adds and one possible impact worth measuring.

MODULE 01

Mission: grow more with less

Define success before choosing a box



Figure 3. One mission can produce different ideas. Criteria and evidence—not first impressions—decide which idea fits best.

Farm in a Box asks you to design a compact crop-production system that uses resources thoughtfully. The challenge is agricultural: support growth, observe the system and improve decisions. An attractive container is useful only if it also supports that purpose. “Grow more with less” means examining the relationship between useful outputs and the resources, space and time involved.

Begin with a problem statement. It should describe the need, the user or situation, and an important boundary without choosing the final solution. “Build a wick box” selects the answer too early. A more open starting point is to support crop production within a limited space while using resources carefully. That leaves room to compare genuinely different ideas.

Criteria describe success. They need to be observable or measurable, such as fitting the allocated space, allowing growth records to be collected, or making avoidable water loss visible. Constraints are actual limits: time, approved resources, available space and opportunities for care. A limit can focus creativity by making the team decide which features matter most.

Teacher confirmation is still required for the crop, dimensions, materials and local procedures. Do not turn an illustrative idea into an approved plan. Later, evidence should let you judge the design against the criteria, including what needs improvement when growth is delayed or unsuccessful.

Write one checkable criterion and one constraint for a compact growing system. Label which is which.

MODULE 01

Apply your learning

Select one best answer for each question. Circle your choice.

01 Which comparison is supported by the named NSW industry sources?

- ☐ A. Rice and cotton produce the same raw material and use identical processing.
- ☐ B. Riverina–Murray rice is a grain industry shaped by irrigation and water availability; NSW cotton is a fibre industry affected by seasonal and water conditions.
- ☐ C. Neither industry is affected by place, season or available resources.
- ☐ D. Rice is mined while cotton is manufactured before it is grown.

02 Which question best examines both benefit and impact at the packaging stage?

- ☐ A. Was the crop harvested?
- ☐ B. Does it protect the product, and what material or waste does it add?
- ☐ C. Can all information be removed?
- ☐ D. Is it the brightest colour?

03 Which interpretation of “grow more with less” is most defensible?

- ☐ A. Choose the smallest-looking idea without testing it
- ☐ B. Compare useful performance with the inputs, space, time and waste involved
- ☐ C. Maximise output even if waste and resource use rise sharply
- ☐ D. Use no inputs at all

System detectives

Choose rice or cotton. Write a short chain from plant to useful product. Identify an input, an intended output and a possible waste output. Explain one feedback record that could improve a stage, then write one checkable resource-use criterion for a small crop system.

MODULE 02

First foods: plants, animals and nutrition

Food knowledge has people and place



Figure 4. Publicly shared research remains connected to named knowledge holders, place and purpose.

Aboriginal and Torres Strait Islander Peoples have diverse, continuing knowledge systems connected to particular Peoples and Countries. A public classroom source shares only part of that knowledge. Responsible learning names who shared it, the place or community involved, what the source actually says and what remains outside our permission or evidence.

A CSIRO public case describes research led with the Wadawurrung Traditional Owners Aboriginal Corporation into the nutritional potential of sovereign grasses. This plant example connects Traditional Owner authority, grassland care and scientific investigation. It does not turn cultural knowledge into freely available instructions for others to copy.

The Gunditjmara public account of Budj Bim Cultural Landscape describes channels and

holding ponds supporting the trapping, containment, live storage and harvest of kooyang, or short-finned eel. This animal-food system on Gunditjmara Country increased food availability. UNESCO corroborates the aquaculture system. The published account supports discussion of management and reliable access, not invented nutrient figures or permission to reproduce practices.

A Queensland Health overview also recognises Torres Strait Islander garden foods and marine animal foods, linking food selection with place, season, energy, health and strong bodies. It broadens the classroom evidence, but is not one island's local Cultural Knowledge. Nutrition and food availability are related questions; do not pretend a source answers both in the same detail.

Name the plant case and animal case. State one claim these public accounts do not authorise.

MODULE 02

Caring for Country and future farms

Compare approaches without blending them

Sustainable resource management involves continuing decisions about land, water, living things and production. The aim is to meet present needs without damaging future capacity. Decisions depend on values and responsibilities as well as observations. Asking who has authority is therefore as important as asking what equipment appears in a photograph.

The publicly shared Wadawurrung sovereign-grasses partnership links Traditional Owner authority, care for grasslands and research. Its meaning remains connected to its named people, purpose and place. One public account cannot represent all Wadawurrung knowledge, and the classroom is not authorised to transfer practices to another Country.

CSIRO's WaterWise case describes sensing, data and prediction used to support irrigation

decisions. Technology can provide information, but people must still interpret it and make decisions. A sensor by itself does not establish that an entire farm is sustainable.

Use common headings to compare the two examples fairly: purpose, decision-maker, evidence, time scale and responsibility. Both connect observation with informed resource decisions. Their knowledge systems, governance, technologies, scale and purposes are not interchangeable. A similarity does not remove those differences. Where a public source leaves a detail unstated, mark it as unknown instead of filling the gap. A careful comparison explains what each source supports without ranking one knowledge system above the other.



Figure 5. Compare purpose, authority, evidence and context. Do not treat distinct approaches as interchangeable.

Give one careful similarity and two important differences between the Wadawurrung partnership and WaterWise. Attribute each example to its named context.

MODULE 02

A sustainability lens

Look beyond the first purchase



Figure 6. Sustainability is a whole-life judgement, not a label attached at the beginning.

A growing system is not sustainable simply because it contains a reused part or a healthy plant. Sustainability asks whether present needs are met while protecting future capacity. It includes what happens when materials are obtained, when a system is made and used, and when parts wear out. A useful judgement follows the whole life of the design.

Consider three connected views. Environmental questions concern water, energy, living systems, materials and waste. Social questions concern access, wellbeing, knowledge, fairness and responsibility. Economic questions concern affordability, durability, time, labour and maintenance. A change may benefit one view while creating a difficulty in another. This is a

trade-off, not necessarily a reason to reject the idea.

Imagine a replaceable water-delivery component. Replacement could keep the rest of the system in use for longer, reducing discarded material. However, access to the part, its cost and the time needed to maintain it still matter. This is an illustrative decision, not proof that a particular component is approved or best.

Convert values into checkable criteria. “Be green” cannot guide a fair judgement. “Allow a worn part to be replaced without discarding the whole system” describes something observable. Record the evidence needed, then explain both the benefit and the remaining burden.

For a replaceable component, identify one environmental benefit and one economic question.

MODULE 02

Apply your learning

Select one best answer for each question. Circle your choice.

01 Which evidence set most directly addresses the required plant-and-animal investigation?

- A. The Wadawurrung sovereign-grasses case, the Gunditjmara Budj Bim kooyang case and a clearly bounded Torres Strait Islander public overview
- B. A local harvesting demonstration invented from the public pages
- C. An unattributed list of foods copied from social media
- D. One photograph with a guessed cultural meaning

02 Which comparison statement is strongest?

- A. Both use evidence, while their knowledge systems, governance and technologies remain distinct
- B. They both care about resources, so every other feature is equivalent
- C. WaterWise replaces the need for cultural knowledge
- D. The grasslands case is an old technology

03 Which judgement uses a complete sustainability lens?

- A. It uses little material, so every impact must be positive
- B. It is cheap to build, so maintenance does not matter
- C. It looks natural, so no evidence is needed
- D. It compares resource effects, people and responsibility, affordability and performance over time

Keep each source visible

Compare the Wadawurrung sovereign-grasses plant case and Gunditjmara Budj Bim kooyang animal case. For each, name the knowledge holders/place and what the public evidence supports about nutrition or food availability. Add how the Queensland Health Torres Strait Islander overview broadens the discussion. Finish with one claim you must leave unmade.

MODULE 03

Plant parts that do the work

A plant is a connected system



Figure 7. Plant structures have different jobs, but their flows connect the whole living system.

Roots, stems and leaves have different jobs, but they depend on each other. Roots usually anchor the plant and take up water and dissolved mineral nutrients. Stems support leaves and reproductive structures, and contain tissues that transport water, minerals and sugars. Leaves provide surfaces exposed to light and air. A problem in one part can affect the whole crop.

Photosynthesis explains an important flow. Plant cells use light energy to build sugars from carbon dioxide and water, releasing oxygen. Carbon dioxide enters from the air; water comes through the plant. Light provides energy, not a material ingredient. The sugars can support

growth, be stored, or move to other parts. Roots do not “eat soil”, and mineral nutrients are not the same thing as these sugars.

Flowers, fruits and seeds have roles in reproduction in flowering plants. The part valued at harvest depends on the crop: it may be a leaf, root, stem, fruit or seed. Knowing the intended product helps a team decide which observations will show quality later.

A plant illustration is a model, not a promise that every species has the same form. Use it to follow connections. For example, damaged roots may reduce the supply of water to leaves, but an observation of poor leaves alone does not prove which cause is responsible.

Sketch a plant from the illustration. Label roots, stem and leaves, and link each to its function. Add a visible reproductive structure only if present.

Explain why a damaged root system can affect leaves, even when the leaves receive light.

MODULE 03

Germination and crop choice

Choose a crop that fits the conditions

Germination begins when a viable seed starts growing. A viable seed contains a living embryo and stored resources. Under suitable conditions, the seed takes up water, its metabolism becomes active, and the young root and shoot emerge. A seed that looks normal is not a guarantee of successful growth.

Most seeds require a suitable combination of water, oxygen and temperature. More of an input is not always better: excessive water can reduce access to oxygen. Light requirements vary between species, so reliable crop information must guide that decision. Avoid the claim that every seed needs exactly the same conditions.

Crop choice should fit the project, rather than simply the most appealing packet picture. Consider the confirmed growing conditions, available time, space, care opportunities and intended product. A short life cycle can be helpful in a school unit, but speed is not useful if the crop does not suit the location or purpose. The teacher confirms the crop and practical arrangements.

Plan observations before starting. Record the date and baseline, decide what will count as emergence, and use a consistent observation method. Keep relevant conditions matched if comparing options. Unexpected or delayed growth still provides evidence when records are honest. A plan should support investigation, not promise a germination percentage or a harvest.



Figure 8. A seed needs suitable conditions, not simply the maximum amount of every input.

Why might the fastest-growing crop still be a poor choice for this project?

MODULE 03

Sow safely

Pause, check, act and record



Figure 9. The safest first move is often a pause: understand the local instruction before beginning.

Safe sowing starts with understanding the actual task. A website or illustration cannot approve a particular tool, growing medium or procedure for your class. Your teacher supplies the current demonstration and controls for the real workspace, equipment and learners. If the instructions are unclear, pause and ask before beginning.

A hazard is something that can cause harm. Risk considers both how likely harm is and how serious it could be. A control reduces the risk. These terms are connected but not interchangeable: naming a hazard does not mean the task is controlled. Students must follow the controls selected for the local situation rather than inventing substitutes.

Use a simple sequence. Pause to understand the task. Check that the workspace is ready and only approved items are present. Act by following the demonstrated method and keeping the work controlled. Stop if conditions change or you become uncertain. Afterwards, leave the area as directed and record what you actually did.

Labels and baseline notes also support responsible production. Crop identity, date and an accurate starting observation help prevent mix-ups and make later comparisons possible. Record teacher observation only when it actually occurred. A completed box on a worksheet is evidence of writing; it is not itself permission to begin practical work.

A classmate wants to substitute an unfamiliar growing material. What should happen before work continues?

MODULE 03

Apply your learning

Select one best answer for each question. Circle your choice.

01 Which statement correctly distinguishes water from sunlight in the plant system?

- ☐ A. Sunlight is a dissolved mineral
- ☐ B. Water is made by roots from light
- ☐ C. Both are material stored as soil
- ☐ D. Water is matter used and moved by the plant; sunlight supplies energy

02 Which crop-choice rationale is strongest?

- ☐ A. Its verified needs and life cycle fit the confirmed conditions, time, space and project purpose
- ☐ B. It is the only plant the student has heard of
- ☐ C. It has the brightest packet picture
- ☐ D. A friend guessed it will grow fastest

03 Conditions change during the task. What is the strongest response?

- ☐ A. Pause, make the area safe as directed and seek the teacher's instruction
- ☐ B. Continue because the original plan can never change
- ☐ C. Delete the record of the change
- ☐ D. Invent a new control privately

Prepare the crop start file

Using a teacher-confirmed crop or supplied example, write its identity and the conditions it needs. Compare it with another approved or supplied crop option. Justify its fit with the project's time, space and care. Plan one dated baseline observation and a consistent follow-up observation. Finish by naming what the teacher must confirm before sowing. If no crop is confirmed, label the entry "proposal".

MODULE 04

What makes a growing medium work?

Balance water, air and support



Figure 10. Roots need water and oxygen. Pore size and connection influence how a medium holds and moves both.

A growing medium surrounds and supports roots. Soil is one possible medium, but some systems use mixtures or other materials. The useful question is not simply “Which looks best?” It is how the medium works with the approved crop, container, water system and care conditions.

Roots need access to water and oxygen. Pores are spaces within the medium. Large connected pores can allow drainage and air movement, but a medium dominated by them may hold less water. Smaller pores can retain more water, yet prolonged saturation can restrict air. Holding the most water is therefore not automatically the best result.

Compare several properties: water-holding capacity, drainage, aeration, support, consistency, mass and nutrient availability. Their importance changes with the context. A light medium may make a portable system easier to move, while creating a trade-off in stability or water storage. A defensible choice names that trade-off instead of hiding it.

Testing gives evidence for particular properties. A drainage observation may show how quickly excess water passes through under the test conditions. It does not prove nutrient suitability or future crop yield. Use the result alongside other evidence, and identify what still needs investigation. The final material choice remains teacher-approved; an attractive photograph cannot establish suitability or authorise its use.

Why is “holds the most water” an incomplete reason for choosing a growing medium?

MODULE 04

Nutrients, compost and circular farming

Follow nutrients through useful pathways

Plants make sugars through photosynthesis, but they also need mineral nutrients for their structures and processes. These are different roles. Mineral nutrients are chemical elements usually taken up in dissolved forms through roots. Adding nutrients does not replace the light, carbon dioxide and water needed to build sugars.

Nutrients move through living and non-living parts of a system. Plants take up available forms; people or animals may use plant products; residues retain material that can enter another process. Decomposers break down dead organic matter and transform it, helping return nutrients to cycling pathways. The material does not simply disappear when its first use ends.

Composting is managed decomposition. Its finished product may return organic matter and nutrients to a growing system. However, not every waste is suitable, and unfinished or unsuitable material is not automatically useful or safe. The teacher must approve local inputs, process controls and use. This reading is not a composting recipe or permission to handle unknown waste.

Circular thinking asks whether an output can safely become a useful input elsewhere. For example, suitable plant residue may enter an approved decomposition pathway. The loop is never perfectly closed: time, labour, energy, losses and quality checks still matter. Evaluate these burdens alongside reduced waste, rather than treating compost as a complete solution.



Figure 11. A circular pathway redirects suitable material, but it still needs controls, time and evidence.

Draw an arrow pathway from suitable plant residue through decomposers to an approved return to the growing system. Label a control and a loss.

Explain one benefit and one limit of returning suitable plant residue through managed composting.

MODULE 04

Run a fair media test

Make a comparison you can trust



Figure 12. Fair comparison comes from consistent conditions, relevant measures and honest repetition.

A fair test changes one selected factor while keeping other important conditions as consistent as practical. This helps a team distinguish the effect being investigated from competing explanations. Living systems still vary, so fairness does not promise identical results or remove the need for careful records.

The independent variable is what you deliberately change, such as the approved growing-medium option. The dependent variable is the response you measure or observe. Controlled variables are conditions kept consistent. If both medium and container size change, a difference in drainage could have more than one explanation.

Write the investigation question before the method. Specify what is measured, when, with

what units, and where the results will be recorded. Apply the same measuring approach to each option. Repeated trials or multiple samples help reveal how variable the result is and whether a pattern is consistent. Record unusual results honestly rather than selecting only those that suit the prediction.

Validity asks whether the test addresses its question. Reliability concerns the consistency of the evidence. A carefully repeated drainage test can inform a drainage criterion, but it does not establish nutrient suitability or future harvest. Use conclusions such as “under these test conditions”, then identify the further evidence needed for the whole design decision. Practical arrangements require teacher confirmation.

In a media drainage comparison, identify the independent variable and one useful controlled variable.

MODULE 04

Apply your learning

Select one best answer for each question. Circle your choice.

01 Why is a drainage test alone insufficient for final media selection?

- A. Evidence should never influence selection
- B. Drainage never affects roots
- C. Every medium drains identically
- D. It does not directly test every property, such as nutrient suitability or long-term support

02 Why is no agricultural loop perfectly closed?

- A. Decomposition stops all nutrient movement
- B. Records make circular systems impossible
- C. No material can ever be reused
- D. Processes still involve time, energy, labour, losses and quality controls

03 How should test evidence influence a design decision?

- A. Choose before testing and rewrite the data
- B. Treat all untested properties as perfect
- C. Ignore criteria and select the result with the largest number
- D. Connect the measured result to a relevant criterion and weigh it with other evidence

Plan the root-zone evidence test

Plan a comparison of two teacher-approved media for one property. State the question, independent and dependent variables, two controlled conditions, measuring units and a repeat. Explain which design criterion the result could inform and one conclusion it could not support. This is a paper plan until the teacher approves the method.

MODULE 05

Where every drop goes

Account for water, not just watering



Figure 13. Water can be stored, taken up, transpired, evaporated, drained or lost as runoff. Design starts by seeing the pathways.

Water entering a growing system can follow several pathways at once. Some stays in the medium, some enters roots, and some drains below the root zone. Water can also evaporate from exposed non-living surfaces or run off when it cannot enter or be contained. A falling reservoir level does not, by itself, identify which pathway caused the change.

Transpiration is water moving through a plant and being released as vapour, mainly from leaves. It is part of normal plant function, so it should not simply be labelled waste.

Evaporation is the change to vapour from non-living surfaces. Distinguishing the two helps designers ask better questions about what a change might affect.

A reservoir level falls. Name two possible pathways and explain why another observation is needed.

A water budget accounts for input, storage, uptake and water leaving by different routes. A classroom budget may be incomplete, but repeated observations can still reveal patterns. Record visible drainage, reservoir changes, root-zone condition and plant observations using a consistent approach. Do not claim precise amounts for pathways that were not measured.

Water-smart management aims to match supply with crop and system needs. Too little can limit growth; poorly controlled excess can also create problems. “Use less” is not a complete criterion. Make loss or shortage visible, relate it to plant condition, and use evidence to decide what to investigate or adjust.

MODULE 05

Choose a water-smart idea

Choose the job before the technology

Water-delivery ideas distribute both water and responsibility. Manual top watering is simple to observe, but relies on regular human decisions. Drip delivery can place water near selected points, but its layout and operation need checking. A passive wick can move water without a pump, although its performance depends on the material, contact, height, medium and conditions.

Compare ideas using common criteria. Consider the root zone, control, visibility, maintenance, energy, cost, reliability and user access. Confirmed care opportunities matter: a system that is easy to operate during a lesson may still need an absence plan. No option removes the need to observe the crop and check for problems.

CSIRO's WaterWise example uses sensing, data and prediction to support irrigation decisions. The transferable lesson is that better observations can improve timing and decisions. It does not mean a school project requires expensive sensors or that an automatic system is automatically sustainable. Extra technology can add cost and interpretation work.

A concept drawing should show the water source, delivery path, root zone, drainage or overflow route, checking access and feedback point. Mark where a failure would become visible. The drawing remains a proposal until the teacher confirms materials and local controls and the prototype is tested. Judge the whole task, not the appeal of the gadget.



Figure 14. Different systems shift control, effort and maintenance. Compare the whole job, not the gadget.

Compare manual watering and a passive wick using the same criterion: maintenance or visibility.

MODULE 05

Build and test a wick

Test the pathway before trusting it



Figure 15. A wick works only when narrow pathways and contact stay connected from source to destination.

A wick can transfer water through narrow connected spaces without a powered pump. Adhesion is attraction between water and another surface, such as a fibre. Cohesion is attraction among water molecules. Together with surface effects, these interactions support capillary action. The wick provides a pathway for existing water; it does not create water.

Transfer has limits. Material structure, cross-section, length, vertical height and moisture conditions can affect performance. Effective contact is needed with the water source, along the pathway and at the receiving medium. A break in that connection may interrupt transfer even when the individual parts look suitable.

A useful prototype changes one approved variable and measures a relevant response over

an agreed interval. Depending on the teacher-approved method, evidence might be visible travel distance, a change in mass or a water-level change. Keep other conditions matched and repeat the test where practical to reveal variation. The exact material and arrangement still require local approval.

If transfer is weak or absent, record the result. Inspect contact and continuity, then choose one controlled modification. Changing every component would make the next result difficult to explain. A successful bench test shows water transfer under those conditions, not guaranteed crop yield. The complete system still needs checking with its actual root zone, care arrangements and conditions.

A wick transfers no visible water. Propose a sensible next step and state what the test cannot prove.

MODULE 05

Apply your learning

Select one best answer for each question. Circle your choice.

01 What is the strongest purpose of a water budget?

- A. Set one universal watering amount
- B. Prove no water ever leaves the system
- C. Ignore plant observations
- D. Account for water entering, being stored or used, and leaving by different pathways

02 Why can “automatic” be a misleading quality claim?

- A. Automated systems still need suitable settings, maintenance, evidence and human oversight
- B. Automation is always cheaper
- C. Automatic systems cannot use data
- D. Automation means the system has no components

03 Which conclusion stays within the evidence?

- A. Capillary action removes all maintenance
- B. This arrangement transferred water during the bench test; full-system performance still needs checking
- C. The test proves no drainage can occur
- D. This wick guarantees ideal water for every crop

Make every drop visible

For a hypothetical compact crop system, compare manual, drip and passive-wick delivery using maintenance and checking access. Select one idea for a prototype and justify the trade-off. Describe one observable failure point and one measurement that would help test it. Keep materials and local care arrangements labelled “Teacher to confirm”.

MODULE 06

Read the microclimate



Figure 16. The best position is not a universal answer; it depends on the crop needs and the pattern of conditions at the site.

A microclimate is the combination of conditions in one small place. Two boxes in the same courtyard may experience quite different light, heat, airflow and rain. A wall can reflect heat; an overhang can block rain; a sheltered corner may stay damp. Design decisions therefore begin with the actual site rather than a general description of the weather.

One observation gives a clue, but repeated observations help establish a pattern. Record the position, time and method as well as the result. Use units for measurements and describe visible changes clearly. For example, an illustrative record might show shade reaching one position earlier on several afternoons. That

pattern is more useful than simply calling the position 'shady'. It still does not prove that shade caused a particular plant response.

Connect each possible design change to the evidence. Moving or turning a portable system might alter exposure, while layout could affect access to light and airflow. Treat these as ideas to investigate within teacher-approved controls. Changing several factors together makes the result harder to explain. A useful site investigation identifies one pattern, explains why it might matter and proposes a consistent comparison. Records or supplied photographs can support this reasoning when a live crop is unavailable.

Why would two readings from one afternoon be weaker than records collected consistently across several days?

MODULE 06

Diagnose crop stress

Drooping leaves, changed colour and slow growth are symptoms: things you can observe. They are not complete diagnoses. Similar symptoms may have different causes, and several conditions may act together. Good crop management begins by describing the evidence before choosing an explanation.

Start with the pattern. Note which plants or parts are affected, when the change appeared and whether it is spreading. Then examine the production record for recent changes and compare the site conditions. An unaffected example can provide a useful comparison when one is available. Rank possible explanations according to this evidence. A plausible explanation remains a hypothesis until further evidence supports it.

For example, if a fictional record shows drooping in one group, the symptom alone cannot establish a cause. The next useful step is to compare that group's conditions and recent records with those of other plants. A teacher-approved test may change one factor and record what follows. Changing many things together would hide which change mattered.

Some findings require reporting rather than testing. An unfamiliar organism, unknown material or rapidly spreading damage should be recorded and reported promptly. Follow the teacher's directions about separation and handling; do not guess a treatment. Even a failed crop can provide useful evidence for diagnosis and redesign.



Figure 17. Describe the pattern before naming a cause.

Write one symptom a student can observe and one question that should be investigated before naming its cause.

MODULE 06

Biosecurity and responsible choices



Figure 18. Do not turn an unknown into a travelling unknown: keep it in place, record it and report it.

Biosecurity means reducing the chance that harmful pests, diseases or contaminated material move into, within or out of a production area. In a school project, responsible action does not depend on identifying every organism. It depends on noticing change, avoiding spread and giving the teacher a useful record.

Pause before moving an unfamiliar item. Observe without unnecessary handling, record where and when it appeared and report what you noticed. Keep material separate only as directed. Do not carry, release, spray or compost an unknown simply to remove it from view. The teacher and approved school process determine the local response.

Consider a fictional leaf mark that has appeared on several plants. A dated photograph and

location note preserve evidence without claiming to identify a disease. Moving affected material to another box could spread an unknown problem and make the original pattern harder to investigate.

Responsible choices extend beyond the crop. Social considerations include people's trust and the effect on other growers. Ethical considerations include care for living things and nearby environments. Legal considerations concern the rules and authorised directions that actually apply; students must not invent these. Slowing down long enough to preserve evidence can prevent a small uncertainty becoming a larger problem. A clear record helps an authorised person decide what should happen next.

An unfamiliar mark appears on several leaves. Give your first response and explain one consequence it helps prevent.

MODULE 06

Investigate crop stress

Select one best answer for each question. Circle your choice.

01 Which record is strongest for comparing conditions over several days?

- ☐ A. Different variables measured at random times with no units
- ☐ B. 'The site seemed fine' written once
- ☐ C. A guess about next month's weather
- ☐ D. The same variable recorded with units at stated times

02 Which evidence could help distinguish between two possible explanations?

- ☐ A. An unrelated crop photograph
- ☐ B. A louder guess
- ☐ C. Removing the record
- ☐ D. A production record showing when the symptom and recent changes occurred

03 What is the main purpose of biosecurity?

- ☐ A. To make every plant look identical
- ☐ B. To replace crop records
- ☐ C. To reduce the movement and impact of harmful pests, diseases or contaminated material
- ☐ D. To let students apply any treatment

Farm detectives — paper case

Fictional case: on two visits, leaves in one box droop while a nearby box appears unchanged. The record does not show recent care or site conditions. An unfamiliar mark is also visible. Use arrows: observation → two possible explanations → evidence needed to distinguish them. Identify the missing records. Finish with the responsible next action; do not prescribe a treatment.

MODULE 07

Turn the brief into criteria



Figure 19. Define what success means before falling in love with a shape.

A design brief explains the need, the intended user or situation, the required result and the limits. Farm in a Box asks for a portable, water-aware crop-production system suited to a small space. It does not prescribe one finished shape. Crops, dimensions, materials, tools and practical controls remain teacher-confirmed details.

Separate requirements from constraints. A requirement describes what the solution must achieve, such as supporting the selected crop. A constraint sets a boundary, such as using approved resources or fitting an approved location. Sorting them reveals questions that need answers before production.

Success criteria turn broad aims into checkable judgements. 'Easy to use' is vague. 'Allows

observations without moving the growing area' identifies a feature that can be checked. Link each criterion to evidence, such as a test record or user observation. Include a numerical threshold only when it has been approved; do not create an arbitrary target to make the criterion appear precise.

Use a balanced group of criteria covering agricultural function, resource use, manageability, communication and safe production. Criteria can compete. More capacity might increase mass, while a protective feature might restrict access. A sound design makes these trade-offs visible and explains its choices. Define success before selecting a favourite shape so the criteria guide the design fairly.

Improve the criterion “easy to monitor” so another student could check it.

MODULE 07

Generate, compare and refine

A concept is a possible way for the system to work. Changing only its colour does not create a genuinely different concept. Useful alternatives vary features such as layout, water movement, access, support, portability or observation. Generate several approaches before deciding which deserves further development.

Annotated sketches make concepts understandable. Labels name parts, arrows show relationships or movement, and brief notes explain decisions. Artistic drawing skill matters less than clear communication. Mark unknown local details as approval points rather than guessing dimensions or resources.

Compare concepts against the same criteria using a decision matrix. Weighting a criterion

gives it greater influence when its importance is justified. Support scores with short evidence notes. An unexplained high number can conceal preference instead of demonstrating performance. For example, an illustrative concept might offer convenient observation but awkward access for maintenance; both features need to appear in the comparison.

The highest total is a starting point for judgement, not an automatic instruction to build. Look for a serious weakness or an assumption that still needs testing. Useful features may be combined into a refined concept where appropriate. Keep the original drawing, mark the change and explain the reason. This visible design trail shows how evidence improved the proposal rather than hiding the earlier thinking.

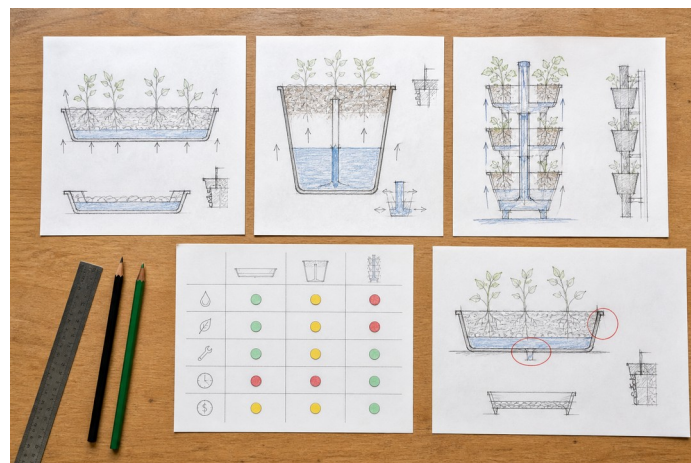


Figure 20. Compare each concept against the same criteria and record the written reason. Counter position supports the comparison; the decision is not conveyed by colour alone.

Why should a concept with the highest matrix total still be examined for weaknesses?

MODULE 07

Plan the build and control risk



Figure 21. The best time to catch a weak step is before the next task depends on it.

A production plan turns an idea into an organised sequence. It identifies tasks, dependencies, evidence and checkpoints. A dependency means that one task relies on something earlier being completed. Some planning tasks may occur together, but a crop should not be established before the system's relevant function and stability have been checked.

A checkpoint is a deliberate pause to compare the work with the drawing, criteria or approved controls. It prevents an early error travelling into later stages. For example, an illustrative plan could require approval of the design before production and a functional check before establishment. The actual sequence and pass conditions must come from the teacher-approved project.

Plan risk controls for the real task. First consider whether an unnecessary hazardous step can be avoided. Then use an approved safer method or resource, organise the workspace and sequence to separate people from hazards, and use instructed personal protection. This planning approach does not approve a tool or procedure: the teacher confirms actual resources, demonstrations and controls. Stop when the work differs from the approved plan.

Allocate roles so work can be coordinated without excluding anyone from the reasoning. A handover note should state what is complete, what was checked and what remains. Everyone needs access to this evidence, even when practical roles rotate.

Name one dependency and explain why a checkpoint belongs before the next task.

MODULE 07

Compare three design concepts

Select one best answer for each question. Circle your choice.

01 Which statement is a constraint?

- ☐ A. The system must work within the approved small space
- ☐ B. The system allows ongoing monitoring
- ☐ C. The idea is creative
- ☐ D. The crop record is understandable

02 What does a weighting represent?

- ☐ A. The relative importance of a criterion
- ☐ B. The physical mass of the design
- ☐ C. A hidden bonus for a favourite concept
- ☐ D. The number of sketches

03 Why place a checkpoint before establishment?

- ☐ A. To make the plan longer
- ☐ B. To check basic function and stability before later work depends on them
- ☐ C. To let students change tools without approval
- ☐ D. To avoid recording evidence

Design studio — justify a refinement

Draw a third, functionally different concept here. Compare it with concepts A and B on page 45 using the same two criteria and weights. Explain the trade-off, chosen refinement and approval still needed.

Concept C with functional annotations

Concept	Criterion 1 × weight	Criterion 2 × weight	Total
A			
B			
C			

Why these weights? Which trade-off and refinement does the evidence support?

MODULE 08

Ready the workspace



Figure 22. Ready means the people, process, place and evidence agree.

A workstation is ready when the people, plan, resources and work area agree. Tidiness alone is not enough. Before production, make the current drawing and sequence available, confirm the immediate task and understand the teacher's directions and stop signal. Use the approved resources and controls for this particular activity.

A readiness gate has three possible decisions: ready; correct and recheck; or stop and ask. It should examine the real task rather than reward a neat appearance. Check that resources match the demonstration, the sequence remains valid and the workspace avoids unnecessary obstruction or mixing of materials. Actual equipment, personal protection and controls are teacher-confirmed.

A bench is tidy but a resource differs from the demonstrated one. Which gate decision is justified, and why?

Keep the drawing, production record and approval information accessible without unnecessary movement through the work zone. If an item is missing, damaged, unfamiliar or different from the plan, do not quietly substitute another. Record the variation and seek direction. For example, a clean bench with an unapproved replacement resource has not passed the readiness gate.

Short, rotating roles help a group work together. One student can examine the space, another read the current step and another record the checkpoint. These roles organise action; they do not remove anyone's responsibility to understand it. Record the decision and the correction or confirmation needed before production continues.

MODULE 08

Build and test the box

Controlled production follows a cycle: complete the approved step, compare the result with the drawing, test the relevant function, record evidence and decide what happens next. The next decision may be to continue, correct and re-test, or stop and seek direction. Recording a correction is part of good production, not a reason to hide the work.

Test function before judging appearance. Depending on the approved design, a test may concern stability, containment, access, portability or the path that resources follow into the growing area. The teacher-approved plan supplies the test method and pass condition. An attractive finish cannot replace the system's main crop-support function.

If a checkpoint reveals a weakness, identify its likely source. Investigate one approved change where possible and repeat the same test. Comparable before-and-after evidence helps show whether the change affected performance. Changing several features together makes that conclusion less certain. Stop if the cause is unclear or the project has departed from the approved design.

Photographs need a useful viewpoint and context. Pair an image with a note explaining the function tested, the result, the decision and the confirmation required for the next stage. A close-up without explanation may look impressive but give another person little evidence to judge. Keep unexpected results as carefully as successful ones.



Figure 23. A checkpoint turns making into controlled production.

What must a before-and-after test keep consistent, and why?

MODULE 08

Establish the crop



Figure 24. A good baseline records the start; it does not pretend to know the finish.

Establishment is the stage when approved planting material enters the working system and its production record begins. The aim is a traceable starting point, not a race to show growth. Later comparisons are fairer when students know what was present and what was done at the beginning.

The teacher confirms the crop, growing medium, tools, handling method, placement, water and other local controls. Follow the demonstrated method and record the actual process. If supplied material or conditions differ from the plan, stop and record the variation before continuing. A workbook illustration is not permission to copy a particular planting arrangement.

Build a baseline record containing the date or project stage, material or supplied sample, position in the system, initial observable condition and approved process completed. An overview photograph shows arrangement; a detail photograph records a feature more closely. Label both so another person can connect them to the written record.

A baseline describes the start; it cannot prove successful future growth. Biological variation means seeds or plants may respond differently even within a project. Delayed growth, failure or interrupted access can still be analysed honestly. Where a live crop is unavailable, teacher-provided photographs or data can support the same comparison skills. Identify supplied evidence clearly and never present it as your own observation.

Why does an establishment photograph need a date, position and written note?

MODULE 08

Control a production checkpoint

Select one best answer for each question. Circle your choice.

01 What are the three possible results of a readiness gate?

- ☐ A. Build, harvest or sell
- ☐ B. Ready, correct and recheck, or stop and ask
- ☐ C. Win, lose or draw
- ☐ D. Rush, guess or hide

02 Who defines the exact functional test and pass condition?

- ☐ A. The most expensive component
- ☐ B. The teacher-approved plan for the real design
- ☐ C. A random online project
- ☐ D. The group after seeing the result

03 Which information belongs in baseline evidence?

- ☐ A. An unrelated crop photograph
- ☐ B. A guaranteed harvest claim
- ☐ C. Only the project title
- ☐ D. Date or stage, supplied material, location, initial condition and completed process

Production traveller — make the decision

Fictional checkpoint: the approved functional test does not meet its stated pass condition. Record the function tested, the evidence you would preserve and the decision to stop or seek approval for one controlled change. Describe how a re-test could be comparable and what must be confirmed before crop establishment. Do not invent a tool, pass value or making procedure.

MODULE 09

Build a useful data record



Figure 25. Record the context first; interpret the pattern second.

Useful field data tells another person what was observed, how and when it was recorded, and which system it describes. Choose a variable because it helps answer a production question. A page of unexplained numbers cannot show that relationship.

Give a table clear headings, units where appropriate and dates or project stages. Use the same measurement method for comparisons. Descriptions such as leaf condition can also be useful when a defined scale is applied consistently. Mark a missed observation as missing; do not replace it with a plausible invented value.

A time-series graph usually places time on the horizontal axis and the measured variable on the

vertical axis. Look for a trend, fluctuation or unusual point. Describe the pattern before suggesting its cause. For example, a rise after a modification is an observation; claiming the modification caused it requires other evidence and consideration of changing conditions.

The course's CSIRO WaterWise example combines field sensing, weather forecasts and data analysis with biological indicators such as canopy temperature to inform irrigation decisions. Sensors do not make sound decisions alone. Crop, soil, climate, water availability and risk still matter. Your manual school record applies the same principle: collect relevant evidence, interpret it in context and keep the conclusion within what the record supports.

A graph rises after a change. What can you say confidently, and what remains uncertain?

MODULE 09

Maintain, test and improve

Maintenance keeps a production system able to perform its intended function. It combines regular observation, approved care, system checks and accurate records. It is more reliable than reacting randomly whenever a plant looks different.

Use a trigger–action–check record. The trigger is evidence that needs attention, such as a repeated observation pattern. The action is the teacher-approved response. The check is later evidence showing whether the issue changed, remained or needs further investigation. Without the final check, an action has been recorded but its effect has not been evaluated.

An improvement begins with a criterion or production problem. Explain the proposed change and why it could help, obtain approval

and alter one factor where possible. Keep the before-and-after method consistent. For an illustrative investigation, changing the observation-access arrangement would need a comparison using the same task and recording method. Any unexpected result belongs in the record too.

If performance improves, the evidence may support that change in this context. It does not prove it will work in every system or season. More technology can produce more data, but extra readings are useful only when they answer the question. The producer still judges data quality, resource use and wider effects. When uncertainty remains, recommend the next evidence step rather than claiming a guaranteed solution.



Figure 26. A useful change has a reason, approval and a re-test.

What is missing from “We changed the system and finished”?

MODULE 09

Grow for community, waste less



Figure 27. Community impact depends on the connections, not just the crop.

A community food initiative connects people, skills and resources. Its contribution might be food production, shared learning, distribution or keeping suitable resources in use. Calling a project 'community' does not prove that it is fair or sustainable; investigate how the system actually works.

Map who contributes, who benefits, what enters and what leaves. Include products, learning, surplus and unusable material. This can reveal a missing connection, such as unclear distribution or too much work falling on one volunteer. Social questions concern access, inclusion and trust. Ethical questions concern responsible treatment of people, living things and environments. Check legal and safety requirements through authorised sources rather than guessing.

Use a teacher-supplied current community growing initiative. Record its name, source and date; map actors, beneficiaries, inputs and waste. Propose a source-backed school connection and one measurable waste-prevention decision. A hypothetical map is preparation only.

Use the waste hierarchy to guide improvement. First prevent avoidable waste through planning and reliable records. Next keep useful resources in use where an approved pathway exists. Recover value through an appropriate authorised process before treating disposal as the last option. Do not assume produce can be eaten, donated or composted: food-safety, allergy, quality and site controls may determine those pathways.

An illustrative initiative might improve its planning by recording repeated unused inputs and adjusting future provision through an approved process. Judge the change using the same records afterwards. Even a small project can contribute useful knowledge, but its claimed impact should match the evidence.

Why is “we will donate every harvest” an unsuitable automatic plan?

MODULE 09

Graph and challenge a claim

Select one best answer for each question. Circle your choice.

01 How should a missing observation be recorded?

- A. Delete the whole table
- B. Mark it honestly as missing or not observed
- C. Copy the previous value without noting it
- D. Invent a likely value

02 What makes a maintenance trigger defensible?

- A. It is based on a rumour
- B. It is linked to a recorded pattern, criterion or system check
- C. It guarantees success
- D. It appears after the action

03 Which is mainly a social consideration?

- A. Whether participation and access are fair
- B. The colour of a graph line
- C. The file extension of a photo
- D. The number of leaves on one plant

Improvement proposal — evidence first

Use a real class record or a labelled teacher-supplied record. Plot one variable against time or condition; label axes, units and scale. Below, state a pattern, a limit, and one approved investigation with a comparable re-check and waste-prevention purpose. If evidence is unavailable, set up labelled axes and request the record; do not invent results.

Graph title and labelled axes

MODULE 10

Harvest and trace the product



Figure 28. Trace what happens at each stage; do not assume every stage applies.

Harvest evidence can be recorded without eating the produce. Use the teacher-approved method to identify the sample, date or stage, quantity or quality, and losses or material remaining in the system. Include units and context. Teacher-supplied photographs or data can support analysis when no live harvest is available.

A supply chain may connect growing, harvesting, processing, packaging, distribution and use. Not every product passes through every stage. A fresh product may need little processing, while another is substantially transformed. Each stage can add value but also use resources, introduce delays or create waste. Separate confirmed school evidence from a researched commercial example.

Compare technologies by the evidence they provide. Current traceability records may identify suppliers, customers, dates, quantities and lot or batch numbers. These help follow a product's identity and movement. The course also describes a CSIRO-reported emerging example: battery-free Bluetooth labels with Internet of Things sensors on produce crates record temperature and location. Such data may reveal a temperature event or delay affecting freshness.

Neither system guarantees quality. Missing or inaccurate data and poor decisions can weaken the chain. Ask who uses the information, which quality decision it supports and what it cannot establish. These are commercial research examples, not equipment requirements for your school crop box.

What different evidence can a batch record and a temperature sensor provide?

MODULE 10

Judge quality with evidence

Quality means fitness for the intended purpose, judged against stated criteria. It does not mean perfection, and crop yield is not the only measure. A disrupted biological result can still reveal strong design reasoning, responsible resource decisions and careful monitoring.

Return to the original success criteria. For each, identify relevant evidence such as a functional test, field record, photograph, maintenance entry or user observation. Judge it met, partly met or not yet demonstrated and explain the limitation. Do not rewrite the criterion after seeing the result simply to make the outcome appear stronger.

Separate product quality from system quality. Product quality concerns the agricultural output, judged against the teacher-provided

specification or class method. System quality concerns how well the designed arrangement performs. Weather, biological variation or interrupted access can affect the crop while records still show whether the system supported observation and resource management.

Triangulation means using more than one suitable evidence type to support a judgement. For example, an illustrative claim about observation access could draw on a recorded test and a contextual photograph, rather than a photograph alone. Evidence must relate to the criterion. When the opportunity to test did not occur, 'not demonstrated' is fairer than declaring failure. State what further evidence would increase confidence and use the evaluation to guide the next version.



Figure 29. Match each claim to named criteria and evidence before judging quality. The evidence relationship, not counter colour, carries the decision.

When is “not yet demonstrated” fairer than “failed”?

MODULE 10

Pitch the next version



Figure 30. Credibility grows when the pitch shows what worked, what did not and what should be tested next.

A design pitch is a short argument supporting a decision. Explain the need, your design response, the strongest result, an important limitation and the next proposed change. Credibility comes from evidence and reasoning, not from hiding difficulties.

Use claim–evidence–reasoning. The claim states your judgement. Evidence names the record that supports it. Reasoning connects that evidence to a criterion or stakeholder. An illustrative claim about reduced input use should identify the relevant log, comparison and test conditions. Simply describing a whole system as ‘sustainable’ is too broad to show what was established.

A next-version proposal identifies a problem, the feature to change, the expected benefit and a

test. The future benefit is a hypothesis, not a result already achieved. Local crops, resources, tools, dimensions and controls remain approval points. It can be reasonable to recommend further investigation rather than immediate production.

Select the most useful evidence instead of showing every folio page. One clear visual and a logical sequence can help an audience follow the argument. State a limitation that affects confidence and end with a specific next decision. If a question exposes missing evidence, acknowledge the gap and explain what should be collected. Honest uncertainty gives the next investigation a purpose and makes the pitch more useful.

Turn “the next version will definitely work” into an evidence-led proposal.

MODULE 10

Pitch the next version

Select one best answer for each question. Circle your choice.

01 Which set belongs in a useful product record?

- ☐ A. The project title only
- ☐ B. A guessed future sale price only
- ☐ C. Stage, product or sample, approved method, quantity or quality evidence and loss
- ☐ D. A photograph with no context

02 Which evidence best supports a system-access criterion?

- ☐ A. A guess about another project
- ☐ B. A harvest mass only
- ☐ C. The colour of the title page
- ☐ D. A documented access test and user observation

03 Which claim is most defensible?

- ☐ A. 'The next version cannot fail'
- ☐ B. 'The system is perfectly sustainable'
- ☐ C. 'The input log shows a reduction after the approved modification, supporting one resource criterion in this test context'
- ☐ D. 'Everyone will use this design'

Field-day pitch — next version

Use your original criteria and strongest records to write a short pitch: need and design response; one claim supported by two evidence items; why that evidence matters; one limitation; and a proposed feature change with its expected benefit and test. Distinguish product quality from system quality. Present future benefits as possibilities, not completed results.

PROJECT FOLIO

Site and success criteria

Observe the proposed site before making a claim about what will grow there. Separate what you saw from what still needs checking.

Write your project problem statement: need, intended users and key limit. Keep it open to different solutions.

Record the site, date, light or shade, access to water, security and one change you would expect during the day.

Write at least four balanced criteria. For each, name what you will judge and the evidence you will collect.

Teacher confirmation to record: approved crop, materials, site and care arrangements.

PROJECT FOLIO

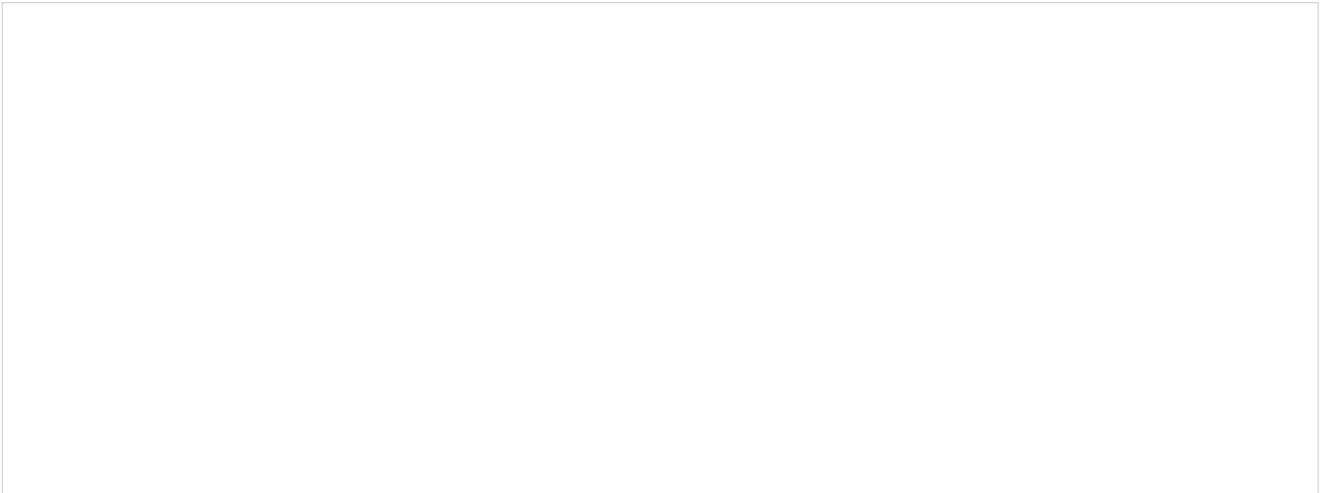
Two concepts and a decision

Change a function, not just the colour. Show the root zone, water route, observation access and portability. Mark unresolved details for approval.

Concept A — annotate how it works.



Concept B — show a genuinely different arrangement.



Which concept best fits your criteria? Explain one trade-off and one refinement.

PROJECT FOLIO

Plan production and control risk

Sequence the work so you can check stability and water movement before establishing the crop. Record only the tools and methods your teacher approves. During production, attach a dated traveller with ready, work, check, actual evidence and next-authorisation entries for each stage. Record teacher observation only when it occurred.

Step and dependency	Approved resources	Check or evidence

Identify two relevant hazards. Record the teacher-approved controls and the point at which you must stop and ask.

Who does what, and how will every person keep evidence of the reasoning?

PROJECT FOLIO

Test one relationship

Complete both a growing-media comparison and a wick-transfer evaluation using approved tests or labelled teacher-supplied records. Use this page for one; attach a second sheet with the same headings for the other. A water-transfer test does not prove crop yield.

Name the test and evidence source. What question, prediction, changed variable, measured response and controls are appropriate?

Trial or date	Condition changed	Result and unit	Observation

What does the evidence support? State a limitation and next check. For media, justify a medium choice; for the wick, justify one change and comparable retest.

PROJECT FOLIO

Establish and care for the crop

Record a baseline when the approved crop is established. A care entry connects evidence, an approved action and a later check. Record absence or missing data honestly.

Baseline: date, crop identity, starting condition and measurement method.

Date	Observation and unit	Approved action	Later check

Care handover: what needs attention before the next visit, and who has confirmed the arrangement?

PROJECT FOLIO

Trace and judge the harvest

Traceability connects the product to its crop, system and handling record. Measuring, grading or photographing the harvest can provide evidence without eating it. Follow the teacher's decision about handling and use.

Record item	Your evidence
Crop and system identity	
Date and maturity evidence	
Harvest or fallback evidence	
Handling and approved destination	

Trace the wider product chain with verbs and arrows. Mark stages confirmed, hypothetical, not applicable or awaiting approval.

Judge two quality features using observations or measurements. What can you conclude, and what remains uncertain?

How could one avoidable input or waste pathway be reduced next time?

PROJECT FOLIO

Evaluate and pitch the next version

A strong pitch joins a claim, a piece of evidence and a reasoned next action. Success includes honest diagnosis and improvement, even when the crop did not reach the intended harvest.

Return to your criteria on page 44. What does your evidence show for each?

Propose one specific change. Why should it help, what approval is needed, and how would you test it?

Prepare a short field-day pitch: need, design choice, strongest evidence and next version.

Course source: Farm in a Box, Wagga Wagga High School. Original workbook explanations and paper activities accompany the course. Course illustrations are illustrative, not approved construction plans.